

## Problem Descriptions

**Exponential distribution.** For a waiting time  $T$  with rate  $\lambda > 0$ ,

$$T \sim \text{Exponential}(\lambda).$$

| Situation    | Equation                                         | Given information | Equation for $\lambda$          |
|--------------|--------------------------------------------------|-------------------|---------------------------------|
| Less than    | $P(T < t) = 1 - e^{-\lambda t}$                  | $P(T > t) = p$    | $\lambda = -\frac{\ln(p)}{t}$   |
| Greater than | $P(T > t) = e^{-\lambda t}$                      | $P(T < t) = p$    | $\lambda = -\frac{\ln(1-p)}{t}$ |
| Between      | $P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$ |                   |                                 |

- Waiting for a TransMilenio bus.** At a TransMilenio station in Bogotá, buses arrive at an average rate of one bus every 5 minutes. Find the probability that the next bus takes more than 6 minutes to arrive.
- Food-delivery requests.** A food-delivery platform in Bogotá receives orders at an average rate of one order every 10 minutes. Find the probability that the next order takes more than 12 minutes to arrive.
- Ride-hailing requests.** A ride-hailing platform in Bogotá receives requests at an average rate of one request every 4 minutes. Find the probability that the next ride request arrives in less than 4 minutes.
- Bank security alerts.** A Colombian digital bank generates security alerts at an average rate of two alerts every 25 minutes. Find the probability that the next security alert is generated in less than 10 minutes.
- Waiting for a study-room opening.** At a university library in Bogotá, study rooms become available at an average rate of one room every 5 minutes. Find the probability that an available room appears between 2 and 7 minutes after the observation time.
- Waiting for a bicycle-repair station.** At a bicycle-repair service in Bogotá, service positions become available at an average rate of three positions every 20 minutes. Find the probability that the next available position appears between 3 and 8 minutes after the observation time.
- Customer-support calls.** A mobile phone company in Bogotá receives technical-support calls throughout the afternoon. The waiting time until the next call is modeled by an exponential distribution. Based on previous observations, there is a 30% probability that the next support call takes more than 8 minutes to arrive. Find the average number of calls received per minute.
- Battery failures.** A company tests rechargeable batteries used in mobile devices. The operating time until a randomly selected battery fails is modeled by an exponential distribution. Based on previous tests, there is a 20% probability that a battery operates for more than 20 hours before failing. Find the average number of battery failures per hour.
- Food-delivery preparation.** A restaurant in Bogotá prepares delivery orders throughout the evening. The preparation time for a randomly selected order is modeled by an exponential distribution. Based on previous observations, there is a 60% probability that an order is ready within 10 minutes. Find the average number of orders prepared per minute.
- Waiting for a bus.** At a Bogotá bus stop, the waiting time until the next bus is modeled by an exponential distribution. Based on previous observations, there is a 75% probability that the next bus arrives within 5 minutes. Find the average number of buses arriving per minute.

# 1 Solutions

## C4.1 - Finding a probability

At a TransMilenio station,  $\lambda = 0.20$  buses/minute. Find  $P(T > 6)$ .

|                |                                  |
|----------------|----------------------------------|
| <b>General</b> | $P(T > t) = e^{-\lambda t}$      |
| <b>Context</b> | $P(T > 6), \quad \lambda = 0.20$ |

Therefore,

$$P(T > 6) = e^{-(0.20)(6)} = e^{-1.2}$$

$$P(T > 6) \approx 0.3010$$

## C4.2 - Finding a probability

The platform wants to find the probability that the next order takes more than 12 minutes to arrive, i.e.  $P(T > 12)$ .

|                |                                   |
|----------------|-----------------------------------|
| <b>General</b> | $P(T > t) = e^{-\lambda t}$       |
| <b>Context</b> | $P(T > 12), \quad \lambda = 0.10$ |

Therefore,

$$P(T > 12) = e^{-(0.10)(12)} = e^{-1.2}$$

$$P(T > 12) \approx 0.3010$$

## C4.3 - Finding a probability

The platform wants to find the probability that the next ride request arrives in less than 4 minutes, i.e.  $P(T < 4)$ .

|                |                                  |
|----------------|----------------------------------|
| <b>General</b> | $P(T < t) = 1 - e^{-\lambda t}$  |
| <b>Context</b> | $P(T < 4), \quad \lambda = 0.25$ |

Therefore,

$$P(T < 4) = 1 - e^{-(0.25)(4)} = 1 - e^{-1}$$

$$P(T < 4) \approx 0.6321$$

#### C4.4 - Finding a probability

The bank wants to find the probability that the next security alert is generated in less than 10 minutes, i.e.  $P(T < 10)$ .

|                |                                   |
|----------------|-----------------------------------|
| <b>General</b> | $P(T < t) = 1 - e^{-\lambda t}$   |
| <b>Context</b> | $P(T < 10), \quad \lambda = 0.08$ |

Therefore,

$$P(T < 10) = 1 - e^{-(0.08)(10)} = 1 - e^{-0.8}$$

$$P(T < 10) \approx 0.5507$$

#### C4.5 - Finding a probability

The library wants to find the probability that an available room appears between 2 and 7 minutes, i.e.  $P(2 < T < 7)$ .

|                |                                                  |
|----------------|--------------------------------------------------|
| <b>General</b> | $P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$ |
| <b>Context</b> | $P(2 < T < 7), \quad \lambda = 0.20$             |

Therefore,

$$\begin{aligned} P(2 < T < 7) &= e^{-(0.20)(2)} - e^{-(0.20)(7)} \\ &= e^{-0.4} - e^{-1.4} \end{aligned}$$

$$P(2 < T < 7) \approx 0.4259$$

#### C4.6 - Finding a probability

The bicycle-repair service wants to find the probability that the next available position appears between 3 and 8 minutes, i.e.  $P(3 < T < 8)$ .

|                |                                                  |
|----------------|--------------------------------------------------|
| <b>General</b> | $P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$ |
| <b>Context</b> | $P(3 < T < 8), \quad \lambda = 0.15$             |

Therefore,

$$\begin{aligned} P(3 < T < 8) &= e^{-(0.15)(3)} - e^{-(0.15)(8)} \\ &= e^{-0.45} - e^{-1.2} \end{aligned}$$

$$P(3 < T < 8) \approx 0.3690$$

### C4.7 - Finding the rate parameter

The company estimates that there is a 30% probability that the next support call takes more than 8 minutes to arrive. Find  $\lambda$ .

$$\begin{array}{l|l} \text{General} & P(T > t) = e^{-\lambda t} \\ \text{Context} & P(T > 8) = 0.30 \end{array}$$

Therefore,

$$0.30 = e^{-8\lambda} \quad \longrightarrow \quad \ln(0.30) = -8\lambda$$

$$\longrightarrow \quad \lambda = -\frac{\ln(0.30)}{8}$$

$$\lambda \approx 0.1505 \text{ calls/minute}$$

### C4.8 - Finding the rate parameter

There is a 20% probability that a battery operates for more than 20 hours before failing. Find  $\lambda$ .

$$\begin{array}{l|l} \text{General} & P(T > t) = e^{-\lambda t} \\ \text{Context} & P(T > 20) = 0.20 \end{array}$$

Therefore,

$$0.20 = e^{-20\lambda} \quad \longrightarrow \quad \ln(0.20) = -20\lambda$$

$$\longrightarrow \quad \lambda = -\frac{\ln(0.20)}{20}$$

$$\lambda \approx 0.0805 \text{ failures/hour}$$

### C4.9 - Finding the rate parameter

There is a 60% probability that an order is ready within 10 minutes. Find  $\lambda$ .

$$\begin{array}{l|l} \text{General} & P(T < t) = 1 - e^{-\lambda t} \\ \text{Context} & P(T < 10) = 0.60 \end{array}$$

Therefore,

$$0.60 = 1 - e^{-10\lambda} \quad \longrightarrow \quad e^{-10\lambda} = 0.40$$

$$\longrightarrow \quad \ln(0.40) = -10\lambda \quad \longrightarrow \quad \lambda = -\frac{\ln(0.40)}{10}$$

$$\lambda \approx 0.0916 \text{ orders/minute}$$

#### C4.10 - Finding the rate parameter

There is a 75% probability that the next bus arrives within 5 minutes. Find  $\lambda$ .

|                |                                 |
|----------------|---------------------------------|
| <b>General</b> | $P(T < t) = 1 - e^{-\lambda t}$ |
| <b>Context</b> | $P(T < 5) = 0.75$               |

Therefore,

$$0.75 = 1 - e^{-5\lambda} \quad \longrightarrow \quad e^{-5\lambda} = 0.25$$

$$\longrightarrow \quad \ln(0.25) = -5\lambda \quad \longrightarrow \quad \lambda = -\frac{\ln(0.25)}{5}$$

$$\lambda \approx 0.2773 \text{ buses/minute}$$